

% AgroBd Project Report

This report documents the AgroBd web application — a Laravel-based agricultural e-commerce platform connecting farmers, retailers and buyers to enable fair pricing and direct commerce. Key features include:

- AgroBd is a web-based platform that connects farmers, retailers and buyers to enable fair pricing and direct commerce. Key features include:
- Farmers and retailers often receive unfair prices because of intermediaries. AgroBd aims to create a marketplace where sellers can get fair prices for their products.

Backend: Laravel 8.x (PHP 7.3+/8.0+)

Frontend: Blade templates, Tailwind CSS, Alpine.js, Livewire

Authentication: Laravel Jetstream and Sanctum

Database: MySQL

Payment Gateway: SSLCommerz

Containerization: Docker

CI/CD: Jenkins

The application follows the Laravel MVC architecture. Controllers handle HTTP requests, Models represent database entities, and Views render the user interface.

``app/Models`` - Eloquent models (Product, Order, User, Category, Brand, etc.)

``routes/web.php`` - Web route definitions for admin and frontend

``resources/views`` - Blade templates and UI components

``database/migrations`` - Schema migration files

Below are the primary models and notable fields inferred from the codebase and migrations.

- Fields: ``id``, ``name``, ``email``, ``phone``, ``address``, ``password``, ``email_verified_at``

- Notes: Implements ``MustVerifyEmail``, uses Jetstream features and two-factor authentication.

- Fields: ``id``, ``category_id``, ``brand_id``, ``product_name``, ``product_slug``, ``product_code``, ``product_quantity``, ``description``, ``price``

- Relations: `belongsTo`Category``

- Fields: (guarded) includes ``user_id``, ``order_status``, ``total``, ``shipping_id``, etc.

- Relations: `belongsTo`User``

Category, Brand, Cart, Wishlist, Discount, Payment, OrderItem, OrderPayment, Rating, Review, Shipping, MyBusiness entity.

Key user flows:

- User registration, email verification, login (Jetstream)

- Product listing by sellers (admin/business)

- Search and browse categories

- Cart and checkout flow with SSLCommerz payment handling

- Order history, order status updates, and payment history

- Ratings and reviews after transactions

Steps to run locally:

```
composer install
```

```
npm install
```

```
cp .env.example .env
```

```
php artisan key:generate
```

```
php artisan migrate
```

```
npm run dev
```

The repo contains a ``Dockerfile`` and CI pipeline (Jenkins) for containerization and deployment.

Unit and feature tests are scaffolded under ``tests/`` using PHPUnit. Developer tools include Faker, Mockery, and Laravel Sanctum.

- Uses prepared statements via Eloquent and Laravel's validation.

- Authentication with Jetstream and Sanctum, email verification enabled.

- Payment integration with SSLCommerz; ensure credentials are stored in ``.env`` and not committed.

AgroBd implements a practical marketplace for agricultural goods using Laravel and modern frontend stacks. For future work, see the ``.env`` file for configuration details.

See ``app/Models/Product.php``, ``app/Models/Order.php``, ``app/Models/User.php`` for model definitions.